

6ELEN018W - Tutorial 3 Solutions to Selected Exercises

Exercises 1, 2, 3, 5

These simply involve writing Python code to implement the problems (or similar problems) seen in the lecture.

You should spend the tutorial session attempting these with the help of your tutor. Make sure that you seek the help of your tutor for anything you do not understand or whenever you are stuck with a task.

Exercise 4

1.

$$A = \begin{bmatrix} a_{00} & a_{01} \\ a_{10} & a_{11} \end{bmatrix} = \begin{bmatrix} 0.9 & 0.1 \\ 0.6 & 0.4 \end{bmatrix}$$

$$P(X_0 = 0) = 0.8, P(X_0 = 1) = 0.2$$

2. $P(X_3 = 0 | X_0 = 1) = a_{11}a_{11}a_{10} + a_{10}a_{01}a_{10} + a_{10}a_{00}a_{00} + a_{11}a_{10}a_{00} = 0.834$

3.

$$P(X_1 = 1) = P(X_1 = 1 | X_0 = 0)P(X_0 = 0) + P(X_1 = 1 | X_0 = 1)P(X_0 = 1) = 0.1 \cdot 0.8 + 0.4 \cdot 0.2 = 0.16$$

$$P(X_1 = 0) = P(X_1 = 0 | X_0 = 0)P(X_0 = 0) + P(X_1 = 0 | X_0 = 1)P(X_0 = 1) = 0.9 \cdot 0.8 + 0.6 \cdot 0.2 = 0.84$$